

Introduction to Cloud Service Models

Cloud computing provides IT resources over the internet. Instead of buying and maintaining physical servers and software, organizations use cloud service providers like Amazon Web Services, Microsoft Azure, and Google Cloud.

There are **three main cloud service models**:

1. **SaaS (Software as a Service)**
2. **PaaS (Platform as a Service)**
3. **IaaS (Infrastructure as a Service)**

These models differ based on **who manages what** (hardware, operating system, applications, etc.).

SaaS (Software as a Service) Explained

Definition:

SaaS provides **ready-to-use software applications** over the internet. Users do not need to install or maintain anything.

Who Manages What?

- Infrastructure → Provider
- Platform → Provider
- Application → Provider
- User → Just uses the software

Examples:

- Google Docs
- Microsoft 365
- Salesforce

Real-Life Example:

Using Gmail in your browser. You don't worry about servers or updates — everything is managed by the provider.

✓ Best For:

- Students
 - Businesses needing ready software
 - Email, CRM, file sharing
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2 PaaS (Platform as a Service) Explained

✓ Definition:

PaaS provides a **development platform** where developers can build, test, and deploy applications without managing servers.

✓ Who Manages What?

- Infrastructure → Provider
- Operating System → Provider
- Runtime → Provider
- Application → User

✓ Examples:

- Heroku
- Google App Engine
- Microsoft Azure App Service

✓ Real-Life Example:

A developer wants to create a website. They just upload the code — no need to configure servers.

✓ Best For:

- Application developers
 - Startups building apps quickly
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3 IaaS (Infrastructure as a Service) Explained

✅ **Definition:**

IaaS provides **virtual machines, storage, and networking**. Users manage the operating system and applications.

✅ **Who Manages What?**

- Infrastructure → Provider
- Operating System → User
- Applications → User

✅ **Examples:**

- Amazon EC2
- Google Compute Engine
- Azure Virtual Machines

✅ **Real-Life Example:**

Renting a virtual server online and installing your own software.

✅ **Best For:**

- IT professionals
- Organizations needing full control

